

ALCHEMY AOE ALLIANCE

**TYPE:
STANDARD
(STND)**



**PROJECT:
N/A**

STND--002

REVISION: R1

General Requirements, Random Maps

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1 - Referenced Documents

This “Standard” does not depend on any components.

2 - Definitions

The following definitions apply:

- *Build Time* – The amount of time required for a single villager to construct a building, prior to application of civilization bonuses.
- *Design-Critical Terrains* – Surfaces introducing different game-play mechanics, defined in Table 2 below:

Class	Walk-Able?	Navigable ?	Buildable ?	Resources ?	Building Bonus Damage?	Examples
ICE	Yes	No	No	No	No	ICE, DLC_ROCK, DLC_BOGLAND
GROUND	Yes	No	Yes	Yes	No	GRASS, DESERT, SNOW
CRACKED	Yes	No	Yes	Yes	Yes	DLC_CRACKED
BEACH	Yes	Yes	Only Walls/ Docks	No	No	BEACH, DLC_WETBEACH, DLC_GRAVELBEACH
SHALLOW	Yes	Yes	No	No	No	SHALLOW, DLC_NEWSHALLOW, MUDDY_SHALLOW
WATER	No	Yes	No	No	No	WATER, MED_WATER, DEEP_WATER
MANGROVE	Yes	Yes	Yes	Yes	No	DLC_MANGROVESHALLOW, DLC_MANGROVEFOREST

Table 2: Design-Critical Terrain Class in Age of Empires II

In the case of terrain masking, these definitions apply to the bottom layer because that is where the terrain properties are inherited from.

- *Forest Tree* – Any tree-class object (including, but not limited to: as oak forest tree, pine tree, palm tree, reed, bush, etc.) placed on forest terrain such that it can be built over only after all its stored wood is removed.
- *Grid Alignment Change* – All objects in AOE2 are either *on-grid* or *off-grid*. Most stationary objects (gaia or otherwise) are *on-grid*, meaning that they can be placed directly adjacent against the edge of the map, without space for units to pass between. Some objects are *off-grid*, and if placed directly against the

map edge would produce a $\frac{1}{2}$ tile space for units to slip through. A *grid alignment change* occurs when an object that is normally *on-grid* becomes *off-grid*, or vice-versa. *Grid Alignment Changes* make maps less intuitive because farms may no longer fit properly around town-centers, and walls may have holes if built against objects not of the same alignment.

- *Neutral* – Any map feature that avoids the origins of player starting locations by at least 30 tiles or 25% of map edge length, whichever is less. This threshold distance may decrease if a creator can demonstrate that the map contains blocking terrains or objects.
- *Nomad* – A map where players do not start with a Town-Center.
- *Player-Assigned* – Any map feature that is not *Neutral*.
- *Standard Starting Resources* – Prior to application of civilization-specific bonuses:
 - For non-nomad maps: 200 of each resource: food, wood, gold, and stone.
 - For *nomad* maps: 200 food, 375 wood, 200 gold, and 300 stone.
- *Rule Changes* – Invisible alterations to game mechanics which cannot be immediately and exactly quantified by players as the match unfolds, but still impact play. For example giving boars twice as much food is not a “*Rule Change*” because players can click on the boar to see how much food it contains. Doubling the rate at which food is gathered from boars is considered a “*Rule Change*”, because such a mechanic is not immediately obvious to players, and cannot be inspected during play.
- *Support* – The map loads properly and within 60 seconds for the given configurations/settings without crashing the game or failing to spawn any players. Such players receive the same number of resources, without bias in terrains, elevations, objects, etc. Does NOT mean that the configuration/settings are recommended by the creator, or predicted to result in an enjoyable or entertaining game. The map will load as requested and everyone will have the same potential for an equally wonderful or terrible experience.

3 - Background

The original iteration of this document was drafted by “old ViridArx” guild members and referred to as “SPC-5”. Released October 26th, 2023, it was used to develop maps for Alchemy League Season 1, and initial iterations of “Space Maps”. In June 2024, “SPC-5” was rewritten for general use by any AOE2 community, with input from the RMS community, and assigned “STND-002” in the new document numbering system.

4 - Overview

This document defines the general characteristics of supplier-designed random map scripts used in Age of Empires II competitions.

5 - File

Players need to be able to load the Random Map successfully, without excessive waiting for the file to transfer in lobby or parse during game start-up.

5.A - Nature

The random map shall be contained in a single text file with .rms extension.

5.B - No XS

The random map shall not reference or be influenced by e(x)ternal (s)ubroutines (XS).

5.C - Size

The total size of the containing file shall not exceed 2,500 kilobytes.

6 - Timing

It is important that viewers of an AOE competition do not become bored and click away before action can begin. Within the first five minutes, players shall be capable of establishing **both** of the following:

- Resource income to sustain constant unit production.
- At least one production building from which those units may be generated.

7 - Player Setup

This section describes the player starting conditions, including a geometric arrangement of starting locations and changes to the game data that appear after the map loads.

7.A - Player Number & Teaming Configurations

Unfinished work should not be exhibited. At a minimum, the random map shall *support* the following common configurations:

- 1v1 on a 120x120 tile map.
- 2v2 on a 168x168 tile map.
- 3v3 on a 200x200 tile map.
- 4v4 on a 220x220 tile map **and** 240x240 tile map.
- 2v2v2v2 on a 220x220 tile map **and** 240x240 tile map.
- “No Teams” or “Free-For-All” style, on map size commensurate with the total number of players (points 1-4), rounded up if that number is odd.

7.B - Random Player Arrangement

Variety and exploration are important aspects of competitive Age of Empires. Therefore at least **one** of the following must be true to ensure sufficient randomness and create the need for player to adapt to scouting information:

1. The map is *nomad*, with town center location selected by the players.
2. Players shall be placed using “random_placement”, or “grouped_by_team”, in combination with “create_player_lands”.
3. Players shall be placed using “direct_placement”, with lands assigned using “assign_to”. In this case, **each** player shall have at least eight unique starting locations that are more than 2% distant in both x and y coordinate from each other. Note that starting locations that could belong to other players are not counted in this variety. For example, if blue starts at [25,25] and red starts at [75,75], then a reversal where blue is at [75,75] and red is at [25,25] is not

regarded as a contributor to starting location randomness, since this effect could be achieved by randomly assigning different colors to the **same location**.

7.C - Positional Fairness

The geometric center for each team's starting locations – if selected by the random map – shall be approximately equidistant from location [50,50] (center of the map).

7.D - Team Positions

If the map is used in a team-game competition, the players will need to have an idea of who will be flank or pocket. When “team positions” is activated in lobby, a clockwise sweep from lowest number shall order the players as follows, in one or two configurations:

1. First Ordering: 1 (blue), 3 (green), 5 (cyan), and 7 (gray) against 2 (red), 4 (yellow), 6 (purple), and 8 (orange)
2. Second Ordering: 1 (blue), 3 (green), 5 (cyan), and 7 (gray) against 8 (orange), 6 (purple), 4 (yellow), and 2 (red)

If a linear pattern is used for player placement then “clockwise” is simply “left to right”.

7.E - Rule Changes

Game mechanics should be intuitive so that players are not discouraged from signing up for custom map competitions. Therefore, the following apply:

7.E.1 - Guard State

The random map shall not invoke a “guard_state” to alter the mechanics or win conditions of the game.

7.E.2 - Terrain State

The random map shall not alter the terrains suitable for the construction of new buildings using “terrain_state” or any equivalent command.

7.E.3 - Game Rules

The random map shall not use “effect_amount” or “effect_percent” to invoke changes to player *starting resources* or general game rules.

7.E.4 - Object Grid Alignments

The random map shall not invoke *grid alignment changes* on any player-controlled objects or gaia objects intended to interact with players.

7.E.5 - Object Rules

The random map shall not use “effect_amount” or “effect_percent” to invoke *rule changes* that impact any player-controlled objects or gaia objects intended to interact with players, with the following exceptions:

- a) The player is only able to control the unit by starting with it. The unit is untrainable.
- b) Each player gains the ability to build up to one thematic building in the Dark Age, meeting the below requirements:
 - 1) If the building is capable of producing villagers, then it shall cost no less than 275 wood and 100 stone, and have *build time* no less than 150 seconds.
 - 2) If the building is capable of producing fishing ships, then it shall cost no less than 150 wood, and have *build time* no less than 35 seconds.
 - 3) If the building is capable of producing military units, then it shall cost no less than 175 wood, and have *build time* no less than 50 seconds.
 - 4) If the building is capable of producing trade units, then it shall cost no less than 175 wood, and have *build time* no less than 60 seconds.
 - 5) If the building is capable of producing herdables (such as sheep, turkeys, etc.), then it shall cost no less than 100 wood, and have *build time* no less than 35 seconds.
 - 6) If the building is capable of providing population space, then it shall cost no less than 25 wood, and 25 seconds of *build time*, for every 5 population provided. E. G. A ten population hut requires 50 wood and 50 seconds to build using a single villager.

- 7) Buildings with multiple capabilities, as listed above, shall have the cost and construction time of the most expensive option, plus $\frac{1}{2}$ the cost and construction time of all other options added. For example, if a rule change would give players the ability to construct a building that can train villagers, military units, and herdables, then the cost of such a building would be $(275 + 175/2 + 100/2)$ wood = 413 wood and 100 stone minimum, with *build time* $(150 + 50/2 + 35/2)$ second = 193 seconds minimum.
- c) Each player gains the ability to train up to two thematic units at an appropriate building in the Dark Age, meeting the below requirements.
- 1) Range shall not exceed 1.
 - 2) If a combat unit, unmitigated damage output (before armor of defending unit) shall not exceed 8 damage per second, per 80 resources of investment and 21 seconds of training time. Unit speed shall not exceed 0.9 tiles/second, and vision shall not exceed 4 tiles. Unit armor shall not exceed 0 melee, 1 pierce, and hit-points shall not exceed half of the unit's total resource cost.
 - 3) If a resource-collection unit, statistics (including training rate), shall not exceed those of their closest standard-rules analog.
 - 4) If a harvestable unit (such as sheep), then it shall yield food and cost no less than $\frac{1}{4}$ that food yield in wood to produce. Training time shall be no less than 25 seconds for every 100 food contained in the unit. Other statistics shall not exceed those of the closest standard-rules analog.

8 - Land Generation

The random map shall present an arrangement of *neutral* lands that are approximately equal in total distance from the geometric center of all team starting locations, illustrated per the example of Figure 8:

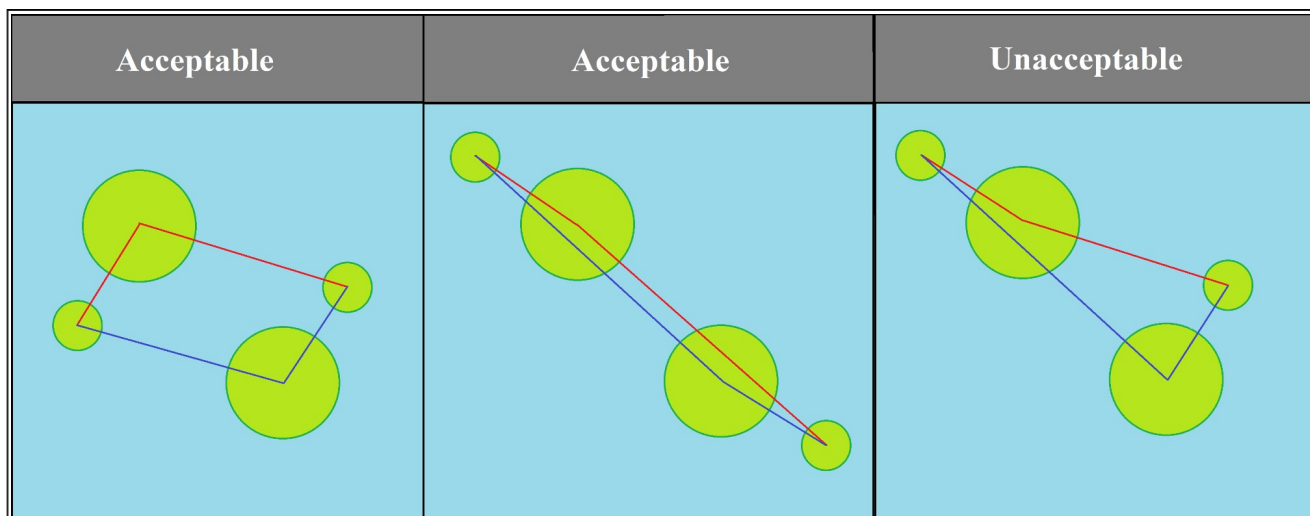


Figure 8: Example of Acceptable and Unacceptable Neutral Land Distribution

For the left and central panes in Figure 8, the total distance of the blue lines and red lines are approximately equal. However, in the right pane, the sum of the blue lines is significantly larger than the sum of the red lines, which gives red an unfair – and unacceptable – advantage.

9 - Elevation Generation

The random map should have a fair distribution of hills, whose heights are not so tall that screens may fail to render them.

9.A - Maximum Height

Maximum Elevation shall not exceed 16.

9.B - Fairness

Elevation shall be uniformly distributed across the random map, without bias toward any edge or team.

10 - Terrain Generation

The random map should have a fair distribution of terrains, per the following:

10.A - Design Critical Terrains

The random map shall not bias any class of *design-critical terrains* in favor of any one team.

10.B - Forest Impingement

With exception of lumbercamps, *forest trees* shall avoid any starting building capable of producing units, being captured, and/or sending/receiving resources by at least 6 tiles.

10.C - Forest Access

10.C.1 - Forest Trees

Players shall have the same number of *player-assigned* forests, each with approximately the same number of trees. Occasional merging of these forest patches is acceptable if overall tree count remains approximately the same.

10.C.2 - Straggler Trees

Non-*nomad* maps shall feature *player-assigned* straggler “tree objects” satisfying **all** of the following conditions:

- Distance 3 to 10 tiles from the origin of player bases.
- Total wood storage of all objects no less than 200.
- No more than 2 per group.
- Less than 10% of all total *player-assigned* wood (including forest trees).

10.D - Forest Readability

Forests shall be indicated on the mini-map in the scenario editor and at the start of the game.

11 - Objects Generation

The random map should have a fair distribution of useful objects, per the following:

11.A - Forbidden Objects

Object 444, known as “Town Center (Packed)” does not work in game as intended and is forbidden.

11.B - Large Eye Candy Readability

Any eye candy object capable of obscuring at least 50% of a player-controlled villager (such as Object 623 “ROCK”) shall be given occlusion mode 2 so that units standing behind it have a visible outline (instead of default concealment).

11.C - Player-Controlled Starting Objects

Objects that start the game under control of players should meet the following requirements:

11.C.1 - Equal Count

With exception of civilization-specific bonuses, the random map shall assure players the same quantity of starting units and buildings, including gaia objects immediately converted as the start of the game.

11.C.2 - Equal Utility

Equal utility of starting objects shall be assured; no starting unit shall be entrapped by forests or other resources, and no starting building shall be pinned against trees or other resources, such that units created from it might be trapped.

11.C.3 - Combat Units

Starting units with the ability to attack or convert cannot be allowed to pose a serious threat to an opponent’s Dark Age economy. This includes units spawned through the deletion of a starting unit. Therefore:

- a) Starting units capable of reaching enemy resource collection, (from being compatible with the same class of terrain) shall have range less than or equal to 1.

- b) On maps where players start with a garrison-able building within 15 tiles of all villagers, the combined attack efforts of all starting military units shall be insufficient to kill a loomed villager (immediately fighting back) at full health, even after achieving “the first hit”.
- c) On maps where players start without a garrison-able building within 15 tiles of all villagers, no starting military unit shall be capable of harming any villagers.

11.D - Gaia Objects

These objects start the game under the control of gaia. They can be resources, friendly, or hostile units/buildings. The following conditions apply:

11.D.1 - Resources

11.D.1.a - Readability

All resources shall be visible through the scenario editor and at the start of the game. This includes resources spawned through the removal of another unit.

11.D.1.b - Player-Assigned

Each player shall receive the same number of *player-assigned* resources, at the same distances from the Town Center.

11.D.1.c - Neutral

Neutral resources shall exhibit no bias toward any team.

11.D.2 - Gaia Convertible Objects

With the following exceptions, the random map shall have no convertible gaia objects:

- a) The object is assured conversion at the start of the game, placed within the vision and capture radius of player-controlled objects.
- b) The object is sheep class (such as sheep, turkey, llama, goat, etc.) such that gaining control of it due to random scouting is not regarded as a large advantage.

- c) The object is a building of approximate equidistance from the team starting locations, that changes control depending on the player who was last closest to it (such as a monument).

11.D.3 - Gaia Hostile Objects

Hostile Gaia Objects are unconvertible and may attack nearby player units or buildings. Gaia hostile objects shall avoid player starting Town Centers by 30 tiles minimum. If players do not start with a garrison-able building within 15 tiles of all villagers, then Gaia Hostile Objects are forbidden.

12 - Regicide Game Mode

The random map shall *support* regicide mode.

13 - Artificial Intelligence

The random map shall be playable by standard AOE2DE artificial intelligence.

14 - Popular Visual Mods

The random map shall *support* the following popular visual mods, without failing to render any terrain or objects:

- 2698 – Age of Cubes Standard Resolution
- 1371 – Simplistic Terrain Colors
- 1205 – Useless Plants Remover
- 790 – Small Trees
- 2695 – Improved Small Trees
- 22014 – Identical Pine Trees with Grid Shadow
- 22781 – Selected Small Trees with Grid Shadow

15 - Trade Configuration

When the “Low Resources” option is selected in lobby, the random map shall disable trade cart and trade cog units and all associated buildings/technologies, at the market and dock respectively, Neither human nor AI players shall be capable of producing such units in that case. No other effect shall be produced from selecting “Low Resources”, (such as decreasing the number of resource piles on the map or giving players a *nomad* start), and no other condition shall determine this *rule change* (such as map size or player count).

Signatures:

TechChariot

2024-06-08

*Prepared**Date (YYYY-MM-DD)*Zetnus

2024-06-11

*Checked**Date (YYYY-MM-DD)*Jasuni

2024-06-11

*Approved**Date (YYYY-MM-DD)*QFilip

2024-06-11

*Approved**Date (YYYY-MM-DD)*

Revision History:

Revision	Description	Change Document	Date (YYYY-MM-DD)
X1	Advanced release for community review/feedback.	N/A	2024-06-08
X2	Incorporated feedback from Zetnus.	N/A	2024-06-08
OI	----- Original Issue -----	N/A	2024-06-12
R1	Updated to new spec format, in addition to other changes: § 2, added definitions for “Grid Alignment”, “Neutral”, and “Player-Assigned”. § 3, added “Background” section. § 7.A, 4th and 5th bullet points: revised FROM “...or...” to “...and...” § 7.E.4, added “Object Grid Alignments”. § 10.C, REMOVED: “Players shall have approximately the same number of forest tress within a 6 to 30 tile radius of any starting Town Center. Simple tress objects within this distance shall not exceed 2 per group, or constitute more than 10% of the unharvested wood within a 30 tile radius from the starting Town Center.” § 10.C.1, added “Forest Trees” section. § 10.C.2, added “Straggler Trees” section. § 10.D, added “Forest Readability” section. § 11.B, added “Large Eye Candy Readability” section. § 11.C.3, point b) revised FROM: “(fighting back)” TO: “(immediately fighting back)” § 11.D, first sentence revised FROM: “These objects start the game as neutral and are under the control of gaia.” TO “These objects start the game under the control of gaia.”	N/A	2025-03-09



R1	§ 11.D.1.a, added “Readability” section. § 11.D.1.b, revised FROM: “Resources that are within 30 tiles of starting Town Centers are “player assigned”. Each player shall receive the same number of such resources, at the same distances from the Town Center. NOTE: <i>Nomad</i> maps tend not to have <i>Player-Assigned</i> resources.” TO: “Each player shall receive the same number of <i>player-assigned</i> resources, at the same distances from the Town Center.” § 11.D.1.c, REMOVED: “Any resources on the map that are not ‘Player-Assigned’ are called ‘Neutral’, and tend toward lower scrutiny by nature of being less relevant to the early game (from being further away).” Section 15, final sentence revised FROM: “No other effect shall be produced from selecting “Low Resources”, such as decreasing the number of resource piles on the map or giving players a <i>nomad</i> start.” TO “No other effect shall be produced from selecting “Low Resources”, (such as decreasing the number of resource piles on the map or giving players a <i>nomad</i> start), and no other condition shall determine this <i>rule change</i> (such as map size or player count).”	N/A	2025-03-09
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